

JEE MAIN REVISION MASTER: CIRCULAR MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 73 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Circular Motion - Core Application	73 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Circular Motion - Core Application (Very High Yield — 73 PYQs)

Core Principle: Key problem pattern in Circular Motion.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A car moving with a speed of 54km/h takes a turn of radius 20 m . A simple pendulum is suspended from the ceiling of the car. Determine th..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30° sh..." [Ans: A]
- [2026 Morning Shift] Q#2 (MCQ): "A smooth inclined plane ends in a vertical circular loop, as shown in the figure. A small body is released from height ■ as shown. If the ..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A car moving with a speed of 54km/h takes a turn of radius 20 m . A simple pendulum is suspended from the ceiling of the car. Determine the angle made by the string of the pendulum with the vertical during the turning. (Take $g=10\text{m/s}^2$)

- (A) $\tan^{-1}(0.5)$
- (B) $\tan^{-1}(0.75)$
- (C) $\tan^{-1}(1.125)$ **[Correct]**
- (D) $\tan^{-1}(0.25)$

Correct Answer: C

Solution Analysis: The speed of the car is given in km/h . We must convert it to m/s to match the units of gravity and radius. $v=54 \times (5 / 18)=3 \times 5=15\text{m/s}$ When the car takes a turn, from the perspective of an observer inside the car, the pendulum bob experiences gravity (mg) which is acting vertically downwards and centrifugal force (F_c) a pseudo-force acting horizontally outwards due to the circular motion. $F_c=\frac{mv^2}{r}$ Where m is mass, v is velocity, and r is radius. Let the string make an angle θ with the vertical. The tension T in the string can be resolved into